



Dynamic Guideline: WHO-Based Rubella - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Current WHO guideline status as of 29 April 2026

- No WHO rubella-specific guideline or update dated 2026 was identified in the provided evidence summaries.
- The most recent WHO rubella vaccine policy document identified was **“Rubella vaccines: WHO position paper – July 2020”** [WHO rubella position paper](#).
- WHO and WHO regional sources after 2020 mainly provide **surveillance standards, laboratory manuals, outbreak-response guidance, immunization implementation plans, epidemiological bulletins, and SAGE updates**, rather than a new rubella treatment guideline.
- The exact title requested in the search scope — **“Meaningful youth engagement in a WHO guideline development process: case study of WHO [] guideline”** — was identified as **“Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)”**, dated 11 December 2025, and is **not related to rubella, measles-rubella, or congenital rubella syndrome (CRS)** [WHO publication B09645](#), [WHO event page](#). It is therefore **not used as a rubella guideline** in this report.

Exactly five WHO or WHO-regional rubella-related guidance documents synthesized

#	Guideline or technical guidance document	Publication or update date identified	Scope and target population	Main relevance
1	Rubella vaccines: WHO position paper – July 2020	July 2020	National immunization programmes; infants, children, adolescents, adults, women of reproductive age	Rubella-containing vaccine policy and prevention of CRS WHO rubella position paper
2	WHO Vaccine-Preventable Diseases Surveillance Standards: Congenital Rubella Syndrome	Last updated 5 September 2018	CRS surveillance systems; infants with suspected CRS; reproductive-age women	CRS surveillance, case detection, and population immunity assessment WHO CRS surveillance standard
3	Manual for the Laboratory-based Surveillance of Measles, Rubella, and Congenital Rubella Syndrome, 3rd edition	2018	Global Measles and Rubella Laboratory Network; public-health laboratories	Laboratory confirmation, quality assurance, specimen testing, and molecular surveillance WHO laboratory manual page

#	Guideline or technical guidance document	Publication or update date identified	Scope and target population	Main relevance
4	Guidelines for measles and rubella outbreak investigation and response in the WHO European Region	2013	Public-health authorities in elimination settings	Integrated measles-rubella surveillance, outbreak definitions, case-based investigation, CRS coordination WHO European Region outbreak guideline PDF
5	PAHO TAG Recommendations for Rubella and Congenital Rubella Syndrome, 1999–2024	PDF covering 1999–2024 recommendations; posted in PAHO source set	WHO Region of the Americas; immunization programmes; surveillance systems	Rubella/CRS elimination, vaccination strategy, integrated surveillance, laboratory network recommendations PAHO TAG recommendations

Guideline Title

For the purposes of current WHO-based rubella guidance, the principal vaccine policy document is:

- **Rubella vaccines: WHO position paper – July 2020** [WHO rubella position paper](#)

Because rubella has **no disease-specific antiviral treatment**, WHO-based management is largely public-health oriented: **prevention through vaccination, case-based and laboratory-supported surveillance, rapid investigation of suspected cases and outbreaks, pregnancy contact management, and CRS surveillance.**

Publication Date

- **July 2020** for the WHO rubella vaccine position paper [WHO rubella position paper](#).
- Related WHO surveillance and laboratory guidance dates range from **2013 to 2018**, with PAHO regional recommendations compiled through **2024**.

Target Population

- **Infants and children** eligible for rubella-containing vaccine.
- **Adolescents and adults** without evidence of immunity.
- **Women and persons of reproductive potential**, especially before pregnancy.
- **Pregnant persons exposed to rubella**, requiring serologic evaluation and follow-up.
- **Infants with suspected CRS**.
- **Public-health programmes, laboratories, immunization managers, and surveillance officers**.

Scope and Objectives

The five WHO or WHO-regional guidance documents collectively aim to:

1. **Prevent rubella and CRS through rubella-containing vaccination.**
2. **Support national introduction and maintenance of rubella-containing vaccine** in immunization programmes.
3. **Detect rubella and CRS through integrated, case-based surveillance.**
4. **Confirm suspected rubella and CRS using laboratory methods**, including serology and RT-PCR.
5. **Investigate and respond rapidly to outbreaks**, especially in elimination settings.
6. **Coordinate rubella surveillance with CRS surveillance**, including pregnancy outcome follow-up.
7. **Maintain laboratory quality assurance and external quality assessment** through the Global Measles and Rubella Laboratory Network.

1.2 Key Recommendations

GRADE note: The provided WHO rubella documents and summaries did **not identify formal WHO GRADE ratings** for rubella recommendations. The ratings below are

therefore **adapted GRADE judgments** based on the summarized evidence, consistency of global guidance, biological plausibility, public-health impact, and strength of evidence. Where evidence is indirect or operational, certainty is downgraded.

Recommendation	GRADE Evidence Quality	Strength
Use rubella-containing vaccine as the primary intervention to prevent rubella and CRS.	High	Strong
Maintain high population coverage with rubella-containing vaccine through routine immunization and catch-up activities where needed.	High	Strong
Use measles elimination and measles-rubella platforms to support rubella vaccine introduction and control.	Moderate	Strong
Do not rely on clinical diagnosis alone for suspected rubella, because symptoms overlap with other viral illnesses and infections may be asymptomatic.	Moderate	Strong
Perform laboratory confirmation of suspected rubella using rubella IgM/IgG serology and, where appropriate, RT-PCR.	Moderate	Strong
For pregnant persons exposed to rubella, collect blood promptly for rubella IgG and IgM , store specimens for possible retesting, and follow pregnancy outcome.	Moderate	Strong
Determine pregnancy status for all persons of childbearing potential who are contacts of suspected or confirmed rubella or CRS cases.	Low	Strong
Conduct CRS surveillance and coordinate it with rubella and measles-rubella surveillance.	Moderate	Strong
Use serosurveys among reproductive-age women as a complementary tool to assess population immunity, but not as a substitute for CRS surveillance .	Low	Conditional

Recommendation	GRADE Evidence Quality	Strength
In elimination settings, implement case-based surveillance and investigate every suspected rubella case and outbreak rapidly.	Moderate	Strong
Integrate measles and rubella surveillance and coordinate rubella surveillance with CRS surveillance.	Moderate	Strong
Maintain laboratory quality assurance, accreditation, external quality assessment, and access to reagents for rubella and CRS testing.	Low	Strong
Use molecular methods, including RT-PCR, to support confirmation and molecular characterization where laboratory capacity exists.	Moderate	Strong
Treat CRS manifestations with multidisciplinary supportive and specialist care; there is no curative antiviral treatment for CRS .	Low	Strong
Ensure postpartum vaccination of persons lacking rubella immunity, consistent with non-WHO authoritative guidance and global prevention principles.	Moderate	Strong

1.3 Guideline Development Methodology

Evidence Review Process

The provided summaries show that the WHO and WHO-regional documents drew on:

- Existing WHO vaccine position papers and surveillance standards.
- WHO laboratory manuals and Global Measles and Rubella Laboratory Network experience.
- Regional outbreak experience, particularly for the WHO European Region outbreak

guideline.

- Expert technical advisory processes, including PAHO Technical Advisory Group recommendations and deliberations of rubella/measles expert panels [PAHO TAG recommendations](#).
- WHO SAGE immunization policy processes; SAGE continued in 2026 to provide transparent, timely evidence-based vaccine recommendations and included rubella among vaccine-preventable diseases with WHO immunization recommendations [WHO SAGE highlights, 18 March 2026](#).

Consensus Method

- The WHO European outbreak guideline was described as based on **existing WHO documents and experience with recent outbreaks in the Region** [WHO European Region outbreak guideline PDF](#).
- PAHO rubella/CRS definitions and recommendations were linked to **expert deliberations**, including the 2004 Ad-hoc Panel of Experts in Rubella and Measles [PAHO TAG recommendations](#).
- The WHO laboratory manual reflects standards for the **Global Measles and Rubella Laboratory Network**, including quality assurance and laboratory surveillance practices [WHO laboratory manual page](#).

Conflict of Interest Management

- The supplied evidence summaries did **not provide detailed conflict-of-interest procedures** for the five rubella-related WHO or WHO-regional documents.
- For WHO vaccine policy generally, recommendations are typically informed by WHO SAGE processes, but no document-specific conflict-of-interest details were available in the provided summaries.

1.4 Clinical and Public-Health Guidance Synthesis

Diagnosis of Rubella

Rubella is usually a mild febrile rash illness, but diagnosis is clinically unreliable. CDC surveillance guidance states that clinical diagnosis alone is unreliable because rubella signs

overlap with other viral illnesses and cases may be asymptomatic [CDC rubella surveillance chapter](#).

Recommended diagnostic approach:

1. Suspect rubella in persons with compatible rash illness, especially if:
 - Unvaccinated or lacking evidence of immunity.
 - Recent travel to an area with rubella transmission.
 - Contact with a confirmed rubella or CRS case.
2. Obtain laboratory confirmation:
 - Rubella IgM and IgG serology.
 - RT-PCR from appropriate clinical specimens where available.
3. Notify public-health authorities promptly.
4. Evaluate pregnancy status in exposed persons of reproductive potential.

Diagnosis and Evaluation of CRS

CRS results from maternal rubella infection during pregnancy. Typical manifestations include hearing impairment, cataracts, congenital heart disease, developmental delay, low birth weight, and other neurodevelopmental complications [CDC CRS surveillance manual](#), [PMC case series](#).

Recommended CRS evaluation:

- Test suspected infants using:
 - Rubella-specific serology.
 - PCR from nasopharyngeal swab, urine, or oral fluid where available.
 - Follow-up serology if needed.
- Assess for:
 - Hearing impairment.
 - Cataracts or other eye disease.
 - Congenital heart disease.
 - Growth restriction.
 - Neurologic and developmental abnormalities.
- Report suspected and confirmed CRS to public-health authorities.

Treatment and Management

There is **no rubella-specific antiviral treatment** and **no cure for CRS**; management is supportive and complication-directed. CDC notes that specific CRS symptoms can be treated, but CRS itself has no cure [CDC pregnancy and rubella](#).

Management principles:

- For uncomplicated rubella:
 - Supportive care.
 - Isolation and public-health control measures according to local guidance.
 - Avoid exposure of pregnant susceptible persons.
- For rubella exposure in pregnancy:
 - Prompt serologic testing.
 - Counseling about fetal risk, especially in the first trimester.
 - Follow-up through pregnancy.
 - Document and report pregnancy outcome.
- For CRS:
 - Multidisciplinary care including cardiology, ophthalmology, audiology, developmental pediatrics, neurology, and rehabilitation.
 - Family support and public-health follow-up to prevent further transmission.

Prevention

Rubella prevention depends on rubella-containing vaccination. WHO and CDC-linked evidence consistently identify vaccination as central to preventing CRS [WHO rubella fact sheet](#), [CDC MMWR 2012–2022](#).

Key prevention actions:

- Maintain high routine immunization coverage.
 - Identify and vaccinate susceptible adolescents and adults before pregnancy.
 - Avoid live rubella-containing vaccination during pregnancy.
 - Vaccinate nonimmune persons postpartum, consistent with CDC best-practice guidance [CDC special situations](#).
 - Use outbreak campaigns and catch-up vaccination where immunity gaps exist.
-

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

The evidence summaries reviewed WHO, WHO regional, CDC, ECDC, NIH-linked/PMC, PubMed, and other public-health sources through **29 April 2026**. Search concepts included:

- Rubella diagnosis.
- Congenital rubella syndrome.
- Rubella-containing vaccine.
- Measles-rubella vaccination campaigns.
- CRS surveillance.
- Laboratory confirmation.
- Pregnancy exposure management.
- Rubella outbreaks and elimination.
- WHO guidance and SAGE updates.

Databases and Sources Searched

The provided summaries included evidence from:

- WHO official pages and PDFs.
- WHO South-East Asia Regional Office epidemiological bulletins.
- WHO SAGE highlights.
- WHO/PAHO technical guidance.
- CDC surveillance manuals and MMWR.
- ECDC CRS information.
- PubMed and PMC articles.
- NCBI Bookshelf/StatPearls.
- Measles & Rubella Partnership publication listings.

No usable rubella-specific clinical trial registry evidence, recent Cochrane review, or recent Lancet, NEJM, JAMA, or BMJ systematic review specific to rubella/CRS diagnosis or treatment was identified in the provided summaries.

Inclusion Criteria

Included evidence addressed at least one of the following:

- Rubella or CRS diagnosis.
- Rubella vaccination or immunization policy.
- Pregnancy exposure management.
- CRS outcomes or burden.
- Surveillance and outbreak control.
- WHO or WHO-regional guidance.
- Laboratory confirmation and molecular surveillance.
- Evidence published or updated after the WHO July 2020 rubella position paper, or relevant later operational updates through 29 April 2026.

Exclusion Criteria

Excluded evidence included:

- Publications not related to rubella, CRS, measles-rubella surveillance, or vaccination.
- The WHO youth-engagement abortion-care guideline case study, because it is not rubella-related [WHO publication B09645](#).
- Non-specific vaccine policy items without rubella-relevant findings.
- Sources without extractable clinical, public-health, or surveillance content.

2.2 Key New Studies

Study or evidence source	Design	Key findings	GRADE Quality
CDC/MMWR global rubella and CRS progress report, 2012–2022	Global surveillance report	Rubella-containing vaccine use expanded globally; reported rubella cases declined from 93,816 in 2012 to 17,407 in 2022; CRS reports increased largely due to expanded CRS surveillance in populous countries	Moderate

Study or evidence source	Design	Key findings	GRADE Quality
		CDC MMWR 2012–2022.	
PubMed-indexed 2025 analysis of estimated CRS incidence in 19 countries, 2019–2055	Modeling analysis	By end-2023, 19 lower- and middle-income countries had not introduced rubella-containing vaccine; a single dose of rubella-containing vaccine can provide lifelong protection; introduction was projected to reduce CRS burden PubMed estimated CRS incidence.	Moderate
CDC continued elimination report, United States, January 2022–June 2024	National surveillance analysis	Measles, rubella, and CRS elimination continued in the United States; rubella IgM has low positive predictive value in elimination settings; updated CSTE confirmed rubella case definition took effect in January 2025 PMC continued elimination.	Moderate
CDC MMWR Florida CRS case, 2025	Field note / case report	Travel-associated CRS can occur despite elimination; clinicians should consider rubella in nonimmune persons with febrile rash illness after travel to endemic regions; South Africa had a large rubella outbreak in 2024 CDC MMWR Florida 2025.	Low
Saskatchewan CRS case series	Case series	CRS occurred in infants born to recently arrived mothers from Sub-Saharan Africa; highlights diagnostic, monitoring, household support, and public-health challenges PMC case series.	Low
CDC rubella	Surveillance	Clinical diagnosis alone is unreliable;	Moderate

Study or evidence source	Design	Key findings	GRADE Quality
surveillance manual	guidance	exposed pregnant persons should receive prompt IgG/IgM testing and pregnancy outcome follow-up CDC rubella surveillance chapter .	
CDC CRS surveillance manual	Surveillance guidance	CRS risk is highest with infection in the first 12 weeks of gestation; congenital anomalies are rare after infection at or after 20 weeks; pregnancy contacts require serologic evaluation, including IgG, IgM, and IgG avidity where indicated CDC CRS surveillance manual .	Moderate
NCBI Bookshelf/ StatPearls 2026 congenital rubella entry	Narrative clinical review	Neonatal CRS diagnosis can use rubella serology and PCR from nasopharyngeal swab, urine, or oral fluid; stable or increasing IgG over the first year supports congenital infection NCBI Bookshelf .	Low
WHO SEARO Regional Vaccine Implementation Plan 2022–2026	Regional implementation plan	Laboratory-supported, case-based measles-rubella surveillance has been initiated in all countries in the region; CRS surveillance is conducted in all countries through case-based or sentinel systems WHO SEARO Regional Vaccine Implementation Plan .	Low
WHO SEARO epidemiological bulletin, April	Operational outbreak bulletin	Bangladesh launched an emergency measles-rubella campaign on 5 April 2026 targeting more than 1.2 million	Low

Study or evidence source	Design	Key findings	GRADE Quality
2026, issue 07		children aged 6 months to 5 years in high-risk districts WHO SEARO Bulletin 2026 issue 07 .	
WHO SEARO epidemiological bulletin, April 2026, issue 08	Operational outbreak bulletin	Bangladesh launched nationwide measles-rubella vaccination on 20 April 2026; 1,595,855 children had been vaccinated by 20 April, representing 73.2% of the campaign target WHO SEARO Bulletin 2026 issue 08 .	Low
WHO Bangladesh Rohingya Crisis Situation Report, March 2026	Humanitarian operational report	Routine immunization continued in Cox's Bazar; 4,120 measles-rubella vaccine doses were administered in March 2026; zero-dose and missed-child identification remained a priority WHO Bangladesh Rohingya Crisis Situation Report .	Low
WHO SAGE highlights, 18 March 2026	Expert advisory update	Rubella remains among vaccine-preventable diseases with WHO recommendations for all immunization programmes; SAGE continues evidence-based vaccine recommendation processes WHO SAGE highlights .	Low
WHO-linked 19th meeting of Measles/ Rubella Regional	Laboratory network meeting report	Emphasized external quality assessment panels, reagent access, outbreak reagent needs, and discussion of whole-genome sequencing for surveillance and	Low

Study or evidence source	Design	Key findings	GRADE Quality
Reference Laboratories		elimination verification 19th meeting report .	

2.3 Evidence Gaps and Future Research Needs

Evidence Gaps

1. No new WHO rubella-specific guideline in 2026

- No WHO rubella guideline or WHO rubella position paper update dated 2026 was identified.

2. Limited formal GRADE reporting

- The provided WHO rubella-related documents did not include extractable formal GRADE evidence profiles.

3. Sparse clinical treatment evidence

- Rubella has no specific antiviral treatment, and CRS management is supportive. Evidence is mainly observational, public-health, or expert-guidance based.

4. Limited high-quality diagnostic accuracy evidence in modern elimination settings

- Low disease prevalence reduces the positive predictive value of rubella IgM testing, creating a need for better diagnostic algorithms in elimination contexts [PMC continued elimination](#).

5. Need for better CRS burden data

- Expanded surveillance has increased CRS reporting, but underdiagnosis and underreporting remain likely in settings without robust CRS surveillance.

6. Limited data in humanitarian and migrant populations

- Recent CRS cases and operational reports highlight risk among internationally mobile, under-immunized, and displaced populations [PMC case series](#), [WHO Bangladesh Rohingya Crisis Situation Report](#).

7. Insufficient evidence on optimal use of genomic surveillance

- WHO laboratory network discussions identified whole-genome sequencing as an emerging area for surveillance and elimination verification, but operational standards

require further development [19th meeting report](#).

Future Research Needs

- Evaluate rubella diagnostic algorithms combining IgM, IgG avidity, RT-PCR, epidemiologic risk, and vaccination history in elimination settings.
 - Improve CRS surveillance sensitivity and specificity in low-resource settings.
 - Assess strategies to identify and vaccinate susceptible reproductive-age populations before pregnancy.
 - Study rubella and CRS risk among migrants, refugees, displaced populations, and zero-dose communities.
 - Evaluate the impact of wide-age-range rubella-containing vaccine campaigns in countries newly introducing rubella vaccine.
 - Develop evidence-based operational guidance for molecular and whole-genome sequencing surveillance.
 - Conduct cost-effectiveness analyses of rubella vaccine introduction in remaining low-income countries.
-

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
Prevent rubella and CRS through rubella-containing vaccination.	Global surveillance and modeling continue to support vaccination as the core CRS prevention strategy; a single dose can provide lifelong protection CDC MMWR 2012–2022 , PubMed estimated CRS incidence .	Maintain a strong recommendation for rubella-containing vaccination in national immunization programmes and catch-up strategies.
Use measles elimination activities to introduce or strengthen rubella	2026 Bangladesh emergency and nationwide measles-rubella campaigns show the continuing operational relevance of combined MR platforms WHO SEARO Bulletin	Use MR campaigns and measles outbreak responses to close rubella immunity gaps, especially in high-risk districts and

Original Recommendation	New Evidence Impact	Updated Recommendation
vaccination.	2026 issue 07 , WHO SEARO Bulletin 2026 issue 08 .	zero-dose communities.
Conduct laboratory-supported case-based surveillance.	WHO SEARO reports that all countries in the region have initiated laboratory-supported case-based measles-rubella surveillance and conduct CRS surveillance WHO SEARO Regional Vaccine Implementation Plan .	Maintain integrated measles-rubella case-based surveillance with laboratory confirmation and CRS surveillance in all countries.
Use serology to evaluate rubella exposure in pregnancy.	CDC surveillance guidance reinforces prompt IgG/IgM testing and specimen storage; CRS guidance adds IgG avidity where appropriate CDC rubella surveillance chapter , CDC CRS surveillance manual .	For every pregnant exposed person, promptly obtain rubella IgG and IgM, consider IgG avidity, store specimens, and follow pregnancy outcome.
Monitor pregnancy outcomes after maternal rubella infection.	Evidence confirms highest fetal risk in early pregnancy and need to document outcomes CDC CRS surveillance manual .	Follow exposed or infected pregnancies through delivery or termination and test infants after birth when indicated.
Confirm suspected CRS through laboratory testing.	StatPearls/NCBI summarizes use of neonatal serology and PCR from nasopharyngeal swab, urine, or oral fluid NCBI Bookshelf .	Evaluate suspected CRS with serology, PCR where available, and longitudinal IgG interpretation, plus full clinical assessment.
Use rubella IgM for diagnosis.	U.S. elimination data highlight lower positive predictive value of IgM in low-prevalence settings PMC	Interpret rubella IgM in context of epidemiologic risk, vaccination history,

Original Recommendation	New Evidence Impact	Updated Recommendation
	continued elimination .	symptoms, travel, exposure history, confirmatory testing, and public-health consultation.
Maintain CRS vigilance even in elimination settings.	2025 Florida CRS case and Saskatchewan cases show importation-related CRS risk CDC MMWR Florida 2025 , PMC case series .	Clinicians should consider rubella/CRS in nonimmune persons with compatible illness or pregnancy exposure after travel or migration from endemic areas.
Strengthen laboratory networks.	WHO-linked lab meeting emphasized EQA, reagent access, outbreak surge capacity, and whole-genome sequencing discussion 19th meeting report .	Maintain laboratory accreditation, EQA participation, reliable reagent supply, molecular testing capacity, and sequencing readiness where feasible.

2.5 Updated GRADE Evidence-to-Recommendation Table

Clinical or public-health question	Evidence summary	Quality of Evidence	Strength of Recommendation	Rationale
Should national programmes use rubella-containing vaccine to	WHO and CDC-linked sources consistently identify vaccination as	High	Strong	Large population impact, consistent evidence,

Clinical or public-health question	Evidence summary	Quality of Evidence	Strength of Recommendation	Rationale
prevent CRS?	the central prevention strategy; global surveillance shows declines in rubella after vaccine introduction WHO rubella fact sheet , CDC MMWR 2012–2022 .			high biological plausibility, favorable benefit-risk balance.
Should countries maintain high routine coverage and catch-up activities?	High coverage and surveillance are described as cornerstones of elimination PMC review .	High	Strong	CRS is severe and vaccine-preventable; under-immunized pockets sustain risk.
Should RCV be introduced in remaining countries without routine rubella vaccine?	Modeling and global reports support introduction to reduce CRS burden; 19 countries still lacked RCV by end-2023 PubMed	Moderate	Strong	Modeled evidence plus strong established vaccine effectiveness and severe preventable outcomes.

Clinical or public-health question	Evidence summary	Quality of Evidence	Strength of Recommendation	Rationale
	estimated CRS incidence.			
Should suspected rubella be laboratory confirmed?	Clinical diagnosis is unreliable; serology and PCR support confirmation CDC rubella surveillance chapter , WHO laboratory manual page .	Moderate	Strong	Accurate diagnosis is essential for outbreak control and pregnancy risk assessment.
Should exposed pregnant persons receive prompt IgG/IgM testing?	CDC recommends prompt blood collection, IgG/IgM testing, and specimen storage CDC rubella surveillance chapter .	Moderate	Strong	High consequence of missed infection; testing guides counseling and follow-up.
Should exposed pregnant persons be monitored through pregnancy?	Asymptomatic maternal infection can cause CRS; fetal risk is highest in first trimester CDC CRS	Moderate	Strong	Follow-up is necessary to identify CRS and document outcomes.

Clinical or public-health question	Evidence summary	Quality of Evidence	Strength of Recommendation	Rationale
	surveillance manual .			
Should CRS surveillance be maintained in all countries?	WHO CRS standards and regional plans emphasize CRS surveillance; reports show CRS case increases with surveillance expansion WHO CRS surveillance standard , WHO SEARO Regional Vaccine Implementation Plan .	Moderate	Strong	Surveillance is required to measure disease burden and vaccine impact.
Should serosurveys replace CRS surveillance?	WHO states serologic assessments can complement but not replace CRS surveillance WHO CRS surveillance standard .	Low	Conditional	Useful for immunity assessment but insufficient to detect CRS outcomes.
Should rubella IgM be interpreted	U.S. elimination evidence shows low positive	Moderate	Strong	False positives can misclassify

Clinical or public-health question	Evidence summary	Quality of Evidence	Strength of Recommendation	Rationale
cautiously in elimination settings?	predictive value of IgM in low-prevalence settings PMC continued elimination .			cases; confirmatory interpretation is needed.
Should suspected CRS receive multidisciplinary clinical management?	CRS causes multisystem disease; no cure exists, but manifestations can be treated CDC pregnancy and rubella .	Low	Strong	Direct trial evidence is limited, but specialist care is clinically necessary.
Should outbreak response prioritize pregnancy contact identification?	CDC surveillance guidance and WHO outbreak guidance emphasize rapid investigation and pregnancy contact assessment CDC rubella surveillance chapter , WHO European Region outbreak guideline PDF .	Low	Strong	High fetal risk justifies strong action despite operational evidence base.

Clinical or public-health question	Evidence summary	Quality of Evidence	Strength of Recommendation	Rationale
Should laboratory networks maintain EQA, reagent access, and molecular capacity?	WHO-linked laboratory reports emphasize EQA, reagents, outbreak surge capacity, and molecular methods 19th meeting report .	Low	Strong	Essential health-system function for elimination verification and outbreak response.

2.6 Practical Clinical Algorithm

A. Suspected Rubella in a Nonpregnant Person

1. Assess symptoms, vaccination history, travel, and exposure.
2. Notify public health if rubella is suspected.
3. Obtain laboratory testing:
 - Rubella IgM and IgG.
 - RT-PCR if available and within appropriate sampling window.
4. Avoid contact with susceptible pregnant persons.
5. Identify contacts and assess immunity.
6. Vaccinate susceptible nonpregnant contacts according to national guidance.

B. Rubella Exposure in Pregnancy

1. Determine gestational age and exposure timing.
2. Obtain blood promptly for:
 - Rubella IgG.
 - Rubella IgM.
 - IgG avidity where available and appropriate.

3. Store serum for repeat testing if needed.
4. Counsel regarding fetal risk:
 - Highest risk in the first 12 weeks.
 - Risk declines later in pregnancy.
 - Congenital anomalies are rare after infection at or beyond 20 weeks, according to CDC CRS guidance [CDC CRS surveillance manual](#).
5. Monitor through pregnancy.
6. Document and report outcome.
7. Test infant after birth if maternal infection occurred or cannot be excluded.

C. Suspected CRS in an Infant

1. Assess for:
 - Cataracts or eye disease.
 - Hearing impairment.
 - Congenital heart disease.
 - Developmental delay.
 - Low birth weight or microcephaly.
 2. Obtain laboratory confirmation:
 - Rubella serology.
 - PCR from nasopharyngeal swab, urine, or oral fluid where available [NCBI Bookshelf](#).
 3. Notify public health.
 4. Provide multidisciplinary management.
 5. Evaluate household and healthcare contacts for immunity.
 6. Implement infection-control and follow-up measures according to local public-health guidance.
-

2.7 Final WHO-Based Dynamic Recommendations as of 29 April 2026

1. **Prevent rubella and CRS primarily through rubella-containing vaccination.**
GRADE: High quality, strong recommendation.
2. **Maintain high routine immunization coverage and close immunity gaps through catch-up or campaign strategies.**

GRADE: High quality, strong recommendation.

3. **Introduce rubella-containing vaccine in any remaining national programmes without it, with appropriate campaign planning and surveillance.**

GRADE: Moderate quality, strong recommendation.

4. **Use integrated measles-rubella surveillance with CRS surveillance in all settings.**

GRADE: Moderate quality, strong recommendation.

5. **Do not diagnose rubella clinically without laboratory confirmation whenever testing is available.**

GRADE: Moderate quality, strong recommendation.

6. **Evaluate all exposed pregnant persons promptly with rubella IgG and IgM, with IgG avidity and repeat testing where indicated.**

GRADE: Moderate quality, strong recommendation.

7. **Follow exposed or infected pregnancies through outcome documentation and infant testing.**

GRADE: Moderate quality, strong recommendation.

8. **Interpret rubella IgM cautiously in elimination settings and incorporate epidemiologic risk, travel, exposure, vaccination history, and confirmatory testing.**

GRADE: Moderate quality, strong recommendation.

9. **Evaluate suspected CRS with laboratory testing and multidisciplinary clinical assessment.**

GRADE: Low quality, strong recommendation.

10. **Strengthen laboratory networks, quality assurance, reagent access, RT-PCR capacity, and molecular surveillance readiness.**

GRADE: Low quality, strong recommendation.

11. **Provide supportive care for rubella and complication-directed multidisciplinary care for CRS; no curative antiviral therapy is available.**

GRADE: Low quality, strong recommendation.

12. **Prioritize susceptible, mobile, migrant, refugee, and zero-dose populations for immunity assessment, vaccination access, and surveillance.**

GRADE: Low quality, strong recommendation.